



Network meta-analysis of knee outcomes following anterior cruciate ligament reconstruction with various types of tendon grafts

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Abstract

Purpose Various grafts were selected for restoring knee stability and regaining pre-injury activity levels after anterior cruciate ligament (ACL) rupture. This review aimed to compare the short-term knee outcomes of different tendon grafts for primary ACL reconstruction (ACLR).

Methods Databases of PubMed, Embase, and CENTRAL were retrieved for identifying the published RCT comparing the short-term (≤ 2 years) knee outcomes of different tendon grafts for ACLR. The main outcomes include the International Knee Documentation Committee (IKDC) subjective and objective scores, side-to-side difference on knee laxity, Lachman test, pivot-shift test, Lysholm score, Tegner score, all recorded complications and graft re-ruptures.

Results A total of 32 trials involving 2962 patients and 17 grafts were included. Superior IKDC subjective score was found for quadriceps tendon autograft (QTB) when compared with autografts including bone-patellar tendon-bone (BPTB), four-strand hamstring tendon (4S-HT), and double-bundle HT. The 4S-HT autograft was associated with poorer anteroposterior and rotational knee stability than the BPTB autograft. Artificial ligament also was found to provide lower knee stability compared with the autografts including the BPTB, patellar tendon, 4S-HT, 4S-HT with preserved tibial insertion, 6S-HT, and four-strand semitendinosus tendon. Additionally, the BPTB allograft showed a statistically higher knee laxity than 6S-HT allograft, while no difference was found between the genuine autografts and non-irradiated allografts.

Conclusions QTB was recommended to be used even over the BPTB and 4S-HT autografts. BPTB autograft was more effective on restoring the knee stability than 4S-HT autograft. Non-irradiated allografts could be safely used as alternatives to autografts. However, artificial ligaments were not recommended for primary ACLR for significantly increased risk of knee laxity.

Keywords Anterior cruciate ligament · Tendon autograft · Tendon allograft · Artificial ligament · Knee function · Knee stability

Xiong-gang Yang and Feng Wang contributed equally to this work.

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Introduction

The anterior cruciate ligament (ACL) tear is one of the most frequently encountered knee injuries in orthopaedic surgery. As such, the incidence of ACL reconstruction (ACLR) has increased over the past few decades [1, 2]. Following the disruption of the ACL, patients typically report symptoms including pain, difficulty in sports, and the knee giving way [3]. Non-surgical treatments have been demonstrated to be associated with suboptimal knee outcomes and often lead to joint deterioration in ACL-deficient patients [4]. Therefore, to adequately restore knee stability and regain pre-injury activity levels, many operative reconstruction managements, including autogenous or allogeneic tendon grafts and artificial

ligaments, have been proposed. However, each graft has its proponents, and orthopedic surgeons continue to debate about which graft is optimal for ACLR.

The central third bone-patellar tendon-bone (BPTB) autograft was firstly introduced in the 1980s for ACLR [5]. It has potential for accelerated bone-to-bone healing and a predictable success rate for restoration of knee stability, for which it was regarded as the “golden standard” in ACLR. However, concerns about the donor-site complications after BPTB autograft harvesting, such as anterior knee pain, crepitus, loss of sensation, patellar fracture, inferior patellar contracture, and loss of extension force, have been proposed in numerous studies [6, 7]. Therefore, the hamstring tendon (HT) autograft has been developed as an alternative to BPTB autograft, resulting in less damage to the extensor apparatus as well as less anterior knee pain and patellar fractures [8]. However, this autograft is also not without shortcomings, as numerous studies have reported that tunnel widening, flexor weakness, and increased knee laxity were more likely to result from the HT than the BPTB autograft [6, 9]. The quadriceps tendon (QT) autograft is another alternative graft of BPTB autograft, which was firstly suggested for ACLR by Marshall et al. [10] in 1979. This autograft has aroused wide concern about the post-operative quadriceps muscle weakness and QT rupture [11]. However, this autograft has recently gained more popularity, as it has demonstrated similar post-operative knee function and stability as well as less donor-site pain when compared with the BPTB and HT autografts [12].

Despite the high success rate in the reconstruction procedures with autograft, surgeons have gradually accepted the use of allogeneic tendons and artificial ligaments, following the advent of some advanced fixation methods and processing technics. All donor site complications would be avoided, and the surgery and recovery time could be shortened. However, the risks of disease transmission, immune rejection, and delayed tendon-bone healing for allogeneic tendon transplantation have become the major challenges for clinicians [13]. Additionally, pre-treatment procedures on the tendon allografts, including both cryopreservation and irradiation sterilization, inevitably cause damage to the mechanical and histological properties, leading to an increased likelihood of re-rupture of the reconstructed ACL [14, 15]. With the use of synthetic ligaments, there are no risks of donor site complications and disease transmission, and fast rehabilitation is permitted [16]. However, the high rate of re-rupture, the depressed ligamentization process, and the increased knee laxity have limited the routine application of such ligaments in clinical practice.

Though many researchers have now compared the treatment outcomes of ACLR with various tendon grafts, disagreements were ubiquitous among the available evidences and there are no directive and informative guidelines, meaning that difficulties remain for surgeons in their choice of an optimal graft. In this study, we will perform a systematic review and network meta-analysis basing on available evidences from randomized controlled trials

(RCTs), with the aim of providing ranking plots for the tendon grafts on the clinical outcomes in a short-term follow-up.

Materials and methods

Study eligibility and selection

The current study was conducted following the guidelines outlined in the Preferred Reporting Items for Systematic Reviews and Meta-analysis (PRISMA) statement [17]. Our systematic review was prospectively registered, and the protocol is available to be read on PROSPERO (registration number CRD42019120661).

RCTs which included patients who had a diagnosis of ACL tears and who had been treated with operative reconstruction using various tendon grafts, comparing short-term (≤ 2 years) knee outcomes, were eligible. Studies were excluded according to the following criteria: (1) duplicated studies; (2) studies which were designed as observational or non-randomized clinical studies, literature reviews, systematic review and/or meta-analyses, case series, case reports or letters to editors; (3) studies without a full-text available. We only included studies which had been published in English.

Two researchers independently conducted database searches on PubMed, Embase, and the Cochrane Central Register of Controlled Trials (CENTRAL) from the inception to November 2018. Detailed searching strategies used in the three electronic platforms are available in Appendix 1.

After initial database searches using these strategies, retrieved studies were checked for duplicate. Following the exclusion of duplicated studies, the titles/abstracts and full-texts of the remaining records were screened for final eligibility. Finally, some other potentially related articles were manually searched after screening the reference sections of head-to-head meta-analyses and other studies.

Data collection and assessment of risk of bias

Two reviewers independently extracted the following data from the included studies: (1) study details: lead author's name, year of publication, study design, location, length of follow-up, and study period; (2) participant details: number of patients, number of participants lost to follow-up, age, gender, injured side, interval between injury and surgery, and meniscal and cartilage status; (3) intervention details: graft type used, fixation methods for both the femoral and the tibial sides, and post-operative rehabilitation program; (4) outcome details: International Knee Documentation Committee (IKDC) subjective score, IKDC objective score, Lachman and pivot-shift tests, side-to-side difference of knee laxity between the operated and contralateral sides by arthrometer (such as KT-1000 or KT-2000), Lysholm score, Tegner score,

overall incidence of complications, and frequency of graft re-rupture. Intent to treatment data was extracted where it was available.

The risk of bias assessment was carried out by two researchers utilizing the Cochrane Collaboration tool [18]. This detects the following seven potential biases for RCTs: randomization sequence generation, allocation concealment, blinding of both participants and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting, and other bias. Each parameter is set as either low risk of bias, high risk of bias, or unclear.

Data synthesis and statistical analysis

We generated network plots for each outcome to illustrate which interventions were compared in available RCTs directly, utilizing the “network” suite of commands for Stata. When outcomes were reported as ordered categorical data, such as the IKDC objective score, Lachman test, pivot-shift test, and side-to-side difference, they were transferred to dichotomous variables as normal (including both normal and near normal grades) and abnormal (including abnormal and obviously abnormal grades) groups, and the odd ratio (OR) was selected as the effect size for further combining, while the mean difference (MD) was used for continuous outcomes. When the outcomes were presented for multiple time points within two years, only the longest follow-up period was selected. In terms of the transferred dichotomous outcomes, a classic half integer continuity correction (adding a 0.5 to each cell) was performed when studies reported zero events in any of the treatment arms.

We conducted all network meta-analyses using the “network” suite of commands in Stata. Prediction interval plots were used to present treatment effect estimates, therefore displaying estimated summary effects with both the 95% credibility interval (95% CI) and the 95% prediction interval (95% PrI). Following this, treatments were ranked according to their estimated effect sizes using the surface under the cumulative ranking curves (SUCRA), showing which treatment ranked the highest, second highest, and so on. Additionally, SUCRA values of the interventions were plotted into radar maps, which coherently displayed treatment effects on various outcome evaluations.

In order to evaluate the inconsistency, the “node-splitting approach” was used to detect the presence of heterogeneity between direct and indirect estimates [19]. Inconsistency plots and heterogeneity variance parameters (τ^2) were used to display the magnitude of inconsistency between the direct and indirect estimates for each loop. Ratio of the two odds ratios (RoR) in the loop were estimated, and the lower 95%CI limit of RoR value close to one was interpreted to mean there was no local inconsistency. We also used a comparison-adjusted funnel plot to assess the presence of small-sample effect in the

network meta-analyses. This study only included a subgroup of patients who had been followed for no more than two years.

Statistical procedures were conducted using the Stata version 14.0 (StataCorp LLC, College Station, TX, USA). Statistical significance was defined as a two-sided *P* value of less than 0.05.

Results

Studies included

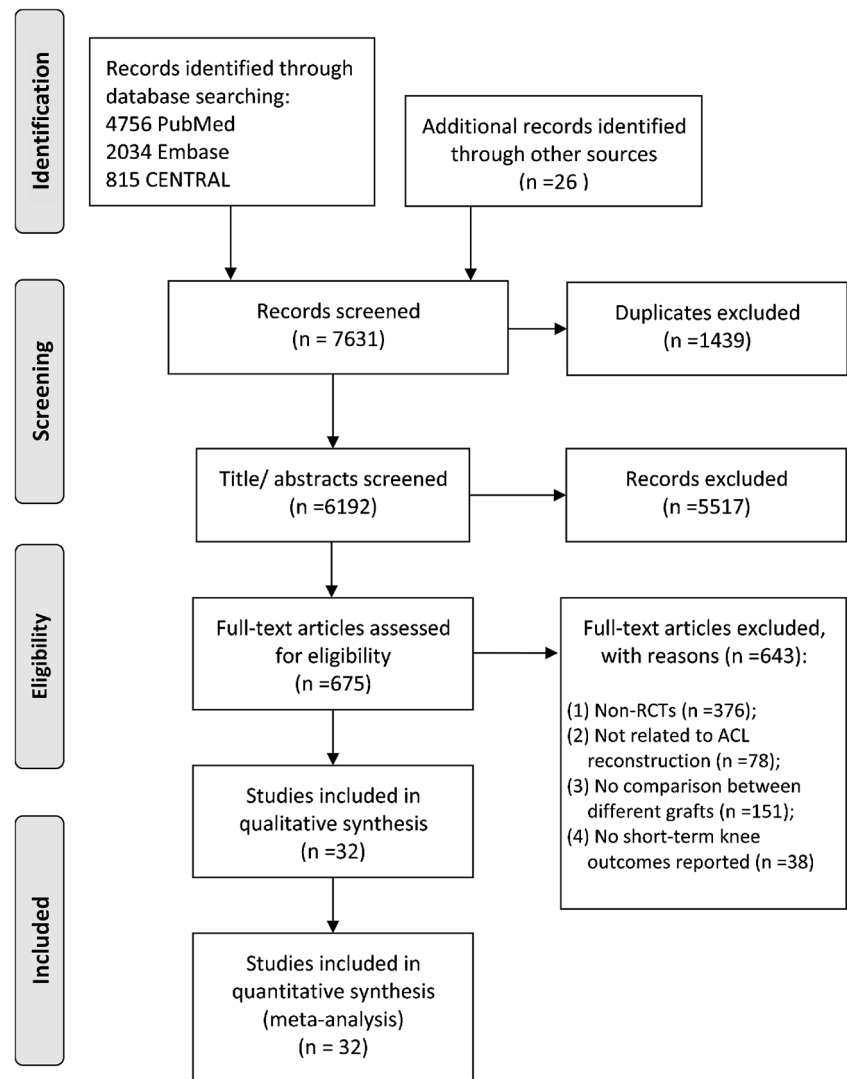
Figure 1 shows the process of searching and selecting studies. The initial retrieval from databases yielded a total of 7605 studies, while 26 potentially eligible studies were manually found, giving a total of 7631 studies. After excluding 1439 duplicates, abstracts and titles of the remaining 6192 studies were screened, while 5517 were excluded as they were not relevant. Following this, 675 articles were considered to be potentially relevant and thus, their full texts were screened. Finally, a total of 32 RCTs [20–51] were included, both for qualitative and quantitative syntheses.

A total of 2962 patients, including 1680 (62.5%) males and 1009 (37.5%) females (the gender of 273 patients was not available), were enrolled. Seventeen different types of grafts were used for ACLR in these patients (Table 1). Table 2 shows the summary of the included studies. Two-arm studies predominated ($n = 30$, 93.8%), with only two studies [41, 43] (6.3%) having three arms. The mean/median age of patients tended to be between 25 and 30 years. The most common treatment arms were the BPTB and four-strand HT autografts, both of which were involved in 22 arms (33.3%). Pre-operative evaluations of the knee function and stability were available in 20 studies [21, 24–27, 30, 31, 33, 34, 36, 38, 39, 41–45, 47, 50, 51] and can be seen in Table S1. Almost all of these comparisons were found to be similar between groups.

Different fixation methods were chosen on the femoral and tibial sides as shown in Figure S1. A summary of the intra-operative meniscal and cartilage status and the rehabilitation protocol are listed in Table S2 and S3, respectively. The application of the protective knee brace was available in 22 studies [20–31, 33, 35, 37, 38, 40, 41, 44–46, 51], eight of which [20, 21, 24, 25, 30, 33, 38, 45] used a brace for protection during the early rehabilitation period.

Summaries of the risk of bias and the risk of bias graph for each study are presented in Fig. 2. In general, the studies included were of a low risk of bias in terms of the random sequence generation, incomplete outcome data, selective reporting, and other bias. However, some studies were found to be have a high risk of bias for allocation concealment, blinding of participants, and personnel as well as blinding of outcome assessment.

Fig. 1 PRISMA flowchart for the searching and selecting of studies



Network meta-analysis of knee function and stability outcomes

The studies, number of patients, and graft types involved in the network for each outcome can be seen in Table S4. Figure 3a–h shows the network plots for knee function and stability outcomes, and the corresponding prediction interval forest plots are presented in Figure S2A–H. There was no evidence that one graft was better than another in terms of the short-term IKDC score, Lachman test, side-to-side difference (categorical data), or Tegner score. Significant comparisons for each outcome are presented in Fig. 4a–d. This shows that the QT autograft with bone block (QTB) has a significantly superior IKDC subjective score than the BPTB autograft, the four-strand HT autograft, and the double-bundle HT autograft, while the four-strand HT autograft has a higher anteroposterior and rotational knee laxity than the BPTB autograft, according to the side-to-side difference (continuous data) and pivot-shift test, respectively. The artificial ligament

was found to be inferior to many other autografts, including the BPTB, PT, four-strand HT, four-strand HT with preserved insertions (HTB), six-strand HT, and four-strand semitendinosus tendon (ST) in terms of the anteroposterior stability according to side-to-side difference, while it was found to be similar to the two-strand HT and three-strand ST autografts. The artificial ligament was found to be superior to the BPTB, BPT, PT, and three-strand ST on the Lysholm score. The BPTB allograft also showed a higher anteroposterior and rotational knee laxity than the six-strand HT allograft according to side-to-side difference and pivot test.

SUCRA ranking plots are shown in Figure S3A–H. We also plotted radar maps depicting the magnitude of SUCRA values for the knee outcomes after they were reconstructed with various tendon grafts (see Fig. 5). When compared with other grafts, patellar autografts (BPTB, BPT, and PT) were shown to have modest short-term knee function outcomes, while the stability is favourable for BPTB autograft but is lowered for

Table 1 Types of the grafts used for ACLR in the included studies

No.	Abbreviated names of grafts	Full names of grafts
1	BPTB-auto	Bone-patella tendon-bone autograft
2	BPT-auto	Patella tendon autograft with bone block on one side
3	PT-auto	Patella tendon autograft without bone block
4	NonIrra-BPTB-allo	Non-irradiated bone-patellar tendon-bone allograft
5	2S-HT-auto	Two-strand hamstring tendon autograft
6	4S-HT-auto	Four-strand hamstring tendon autograft
7	6S-HT-auto	Six-strand hamstring tendon autograft
8	4S-HTB-auto	Four-strand hamstring tendon autograft with Preserved tibial insertion
9	DB-HT-auto	Double-bundle hamstring tendon autograft
10	4S-Contral-HT-auto	Four-strand hamstring tendon autograft harvested from contralateral knee
11	6S-NonIrra-HT-allo	Six-strand non-irradiated hamstring tendon allograft
12	3S-ST-auto	Three-strand semitendinosus tendon autograft
13	4S-ST-auto	Four-strand semitendinosus tendon autograft
14	2S-NonIrra-Achilles-allo	Two-strand non-irradiated Achilles' tendon allograft
15	NonIrra-TA-allo	Non-irradiated tibialis anterior tendon allograft
16	QTB-auto	Quadriceps tendon autograft with bone block
17	Artificial-ligament	Artificial ligament

A total of 17 types of tendon grafts were involved, and this table describes the code, abbreviation, and full name of each graft correspondingly

BPT and PT autografts. When compared with patellar autografts, no obvious differences on knee functional outcomes were found in hamstring grafts, while HT autografts which used no more than four strands were likely to be inferior to patellar tendons on knee stability. However, four-strand HT autografts with preserved bone insertions were associated with better stability than patellar tendons. With the increase on the cross-sectional area of HT autograft, there was a tendency for corresponding elevated anteroposterior and rotational stability. Artificial and allogeneic grafts were generally associated with inferior stability but relatively better short-term knee function.

Network meta-analysis of all recorded complications and re-ruptures

A total of 28 (2645 patients, 16 grafts) of the 32 studies reported data about the number of recorded complications while 22 (1859 patients, 13 grafts) reported data about re-ruptures (Table S4). Network plots and SUCRA ranking plots are shown in Figure S4 and Figure S5. Prediction interval plots for all recorded complications and re-ruptures are presented in Figure S2I–J, while significant comparisons are displayed in Fig. 4E. The double-bundle HT autograft was found to be associated with significantly more complications than the two- or four-strand HT and the BPTB autografts. There was no significant difference in the incidence of re-rupture among the grafts. According to the SUCRA ranking plot, the four-strand contralateral HT autograft had the highest probability of re-rupture. In addition, allogenic grafts (especially for BPTB,

TA, and Achilles) were more likely to suffer re-rupture than autogenous grafts (patellar tendons, QTB, and HT). Patellar tendons had a tendency of lower re-rupture incidence than three- or four-strand ST and four-strand HT, while the six-strand HT and QTB grafts had the lowest re-rupture rate.

Inconsistency assumption and small sample effect

The local inconsistency results for the closed loops are shown in Fig S6, S4C, and S5C. All the lower limits of the 95% CI of RoRs are shown as “1,” and Z tests on the heterogeneity were all non-significant ($P > 0.05$). All local inconsistency detection results based on the “node-splitting model” were also non-significant (Table S5).

The comparison-adjusted funnel plots for knee functional/stability outcomes, complications, and re-rupture events are shown in Figure S7, S4B, and S5B. A subjective symmetry was found, indicating that no obvious small-sample effect exists to increase the risk of publication bias.

Discussion

To our knowledge, this is the first network meta-analysis of various tendon grafts for ACLR that has provided a rank order for those grafts' individual clinical outcomes. The currently published studies have not given a legible tutorial on graft selection for ACLR, meaning evidence is needed for the development of clinical treatment guidelines. Based on the available direct and indirect evidence, a general description for the

Table 2 Summary of included studies

Study ID	Study period	Country	Type of tendon graft	Patients (N)	Dropped patients (N)	Male (%)	Age (years)	Time from injury to surgery	Fixation approach (F = femoral, T = tibial)	Follow-up
Maletis GB, 2007 [20]	2000–2003	USA	BPTB-auto	46	0	67.4	Mean, 27.2 (range 15–42)	< 3 m, 11; 3–6 m, 14; > 6 m, 21	F&T: poly L-lactide bioabsorbable interference screws	Mean, 26.2 (range 23–45)m
Beynon BD, 2002 [21]	1990–1991	USA	4S-HT-auto	53	3	85.0	Mean, 27.7 (range 14–48)	< 3 m, 9; 3–6 m, 12; > 6 m, 32	F&T: interference screw	36 m
			BPTB-auto	28	0	64.3	Mean, 28.5 (range 18–46)	Median, 3.6 (range 0.6–10.2)m	F: a belt-buckle-type double-staple technique; T: staple	
			2S-HT-auto	28	0	46.4	Mean, 29.9 (range 19–42)	Median, 2.6 (range 0.6–9)m	F&T: bioabsorbable interference screw (poly-L-lactic acid screw)	Mean, 19.3 (range 12–25)m
Laoruengthana A, 2009 [22]	2002–2004	Thailand	6S-HT-auto	17	4	88.2	Mean, 29.9	Mean, 35.46 (range 2–108)m	F&T: bioabsorbable interference screw (poly-L-lactic acid screw)	Mean, 18.9 (range 12–31)m
Aune AK, 2001 [23]	N/A	Norway	BPTB-auto	17	2	94.1	Mean, 28.7	Mean, 20.0 (range 3–30)m	F&T: titanium interference screw	24 m
			BPTB-auto	35	6	54.3	Mean, 25.0 ± 7.0	Mean, 41.3 ± 41.0 m	F: titanium button; T: interference screw	
			4S-HT-auto	37	5	56.8	Mean, 27.0 ± 9.0	Mean, 40.5 ± 41.6 m		
Barenius B, 2010 [24]	1995–1997	Sweden	BPTB-auto	84	6	42.7	Mean, 33.0 ± 6.3	Mean, 16.0 ± 24.0 m	F&T: interference screw	Mean, 100.8 ± 11.8 m
			4S-ST-auto	80	5	57.3	Mean, 35.0 ± 7.5		F: EndoButton; T: suture-over-post fixation	
Dai C, 2015 [25]	2007–2009	China	6S-NonIrra-HT-allo	69	8	59.0	Mean, 30.0 ± 6.0	Mean, 10.0 ± 7.0 m	F: EndoButton; T: bioabsorbable interference screw	Mean, 52 (range 30–68)m
			NonIrra-BPTB-allo	60	8	67.3	Mean, 29.0 ± 5.0	Mean, 8.0 ± 6.0 m	F&T: bioabsorbable interference screw	
Drogset JO, 2009 [26]	2001–2004	Norway	4S-HT-auto	57	10	63.5	Median, 27 (range 18–50)	Mean, 12.0 (range 0–240)m	F: Bone Mulch Screw; T: WasherLoc fixation	24 m
			BPTB-auto	58	8		Median, 26 (range 18–49)	Mean, 13.0 (range 2–180)m	F: cannulated interference screw; T: interference screw	
Ejerhed L, 2003 [27]	N/A	Sweden	BPTB-auto	34	2	65.6	Median, 26 (range 14–49)	Median, 11 (range 2–252)m	F&T: Acuflex “silk” interference screw	24 m
			4S-ST-auto	37	3	73.5	Median, 29 (range 15–59)	Median, 17.5 (range 3–360)m	F&T: RCI interference screw	
Feller JA, 2001 [28]	N/A	Australia	4S-HT-auto	31	0	74.2	Mean, 26.2 ± 6.2	Na	F: EndoButton; T: cannulated metallic interference screw	4 m
			BPTB-auto	34	0	70.6	Mean, 27.1 ± 6.2			
Feller JA, 2003 [29]	1996–1998	Australia	4S-HT-auto	31	5	74.2	Mean, 25.8 ± 6.0	Mean, 3.3 ± 3.5 m	F: EndoButton; T: cannulated metallic interference screw	36 m
			BPTB-auto	34	2	70.6	Mean, 26.3 ± 6.0	Mean, 4.6 ± 5.4 m		
	1995–1999	UK		24	2	87.5	Mean, 31.7 ± 1.4		N/A	60 m

Table 2 (continued)

Ghalayini SR, 2010 [30]	Artificial- ligament	PT-auto	26	73.1	Mean, 30.9 ± 1.4	Mean, 55.0 (range 35–75) m	
						Mean, 33.0 (range 19–47) m	NA
Gifstad T, 2013 [31]	Norway	BPTB-auto	58	63.2	Median, 27 (range 18–49)	NA	F&T: metallic interference screws
		4S-HT-auto	56				F&T: Bone Mulch Screw and the WasherLoc
							N/A
Gobbi A, 2006 [32]	Italy	PT-auto	50	N/A	N/A	NA	24 m
		4S-HT-auto	50				
		4S-HTB-auto	55	96.4	Mean, 27.0 ± 7.5	More than 3 weeks	F: EndoButton; T: based on preserved tibial insertions only
Gupta R, 2017 [33]	India	4S-HT-auto	55	94.5	Mean, 27.2 ± 5.7		F: EndoButton; T: bioabsorbable interference screw
							F&T: interference screw
							N/A
Beard DJ, 2001 [34]	UK	4S-HT-auto	30	N/A	N/A	NA	12 m
		PT-auto	30				
		BPTB-auto	34	64.7	Mean, 29.0 ± 7.0	Median, 7.8 (range 5–12) m	24 m
Heijne A, 2010 [35]	Sweden	4S-HT-auto	34	41.2	Mean, 30.0 ± 9.0	Median, 8.5 (range 5–18) m	
Iliopoulos E, 2016 [36]	Greece	BPTB-auto	10	100.0	Mean, 24.8 ± 5.0	NA	F&T: interference screws
		4S-HT-auto	10	100.0	Mean, 26.7 ± 7.2		F: Endobutton; T: interference screw
Jansson KA, 2003 [37]	Finland	BPTB-auto	51	N/A	N/A	Median, 19.0 (range 0.2–240) m	F&T: interference metal screws
		4S-HT-auto	48			Median, 16.0 (range 0.5–120) m	F: small metal plate; T: cortical screw secured with a spiked washer
Kang HJ, 2014 [38]	China	NonIrra-BPTB- allo	43*	46.5	Mean, 30.0 ± 5.0	Mean, 2.1 ± 1.4 m	F&T: interference screws
		NonIrra-TA-allo	41*	51.2	Mean, 28.0 ± 5.0	Mean, 2.3 ± 1.2 m	F: EndoButton; T: bioabsorbable interference screws
							F: cross-pin fixation; T: interference screw
Kautzner J, 2015 [39]	Czech Republic	BPTB-auto	75	0.0	Mean, 26.0 (range 17–47)	Mean, 2.5 (range 0.1–6) m	F: suspension fixation; T: interference screw
		4S-HT-auto	75	0.0			F: cross-pin device; T: washer with multiple spikes and a compression screw
Lawhorn KW, 2012 [40]	USA	4S-HT-auto	74	59.3	Mean, 32.0	Within 12 m	More than 24 m
		NonIrra-TA-allo	73	79.2	Mean, 33.3		
Laxdal G, 2005 [41]	Sweden	BPTB-auto	42	72.5	Median, 28 (range 16–52)	Median, 14.5 (range 2–240) m	Mean, 26.0 (range 20–43) m

Table 2 (continued)

																			F&T: Acufex “silk” interference screw	
Shahab MD, 2002 [42]	1994–1996	USA	3S-ST-auto	41	2	56.4	Median, 24 (range 12–41)	Median, 14.0 (range 2–360) m	F&T: RCI interference screw											
			4S-HT-auto	42	3	71.8	Median, 26 (range 15–35)	Median, 9.0 (range 2–194) m	F&T: RCI interference screw											
			BPTB-auto	33	12	78.8	Mean, 32 (range 14–48)	Mean, 4.6 m	F&T: interference screws											More than 24 m
			4S-HT-auto	37		56.8	Mean, 30 (range 14–53)	Mean, 4.4 m												
Molnadi N, 2015 [43]	2007–2010	Canada	BPTB-auto	110	4	58.2	Mean, 28.7 ± 9.7	NA	F: EndoButton; T: bioabsorbable											24 m
			4S-HT-auto	110	2	53.6	Mean, 28.5 ± 9.9		interference screw											
			DB-HT-auto	110	2	55.5	Mean, 28.3 ± 9.8													
			BPTB-auto	20	0	85.0	Mean, 27.0 ± 5.7	NA	F: two re-absorbable cross pins; T: Rigid-fix and re-absorbable pins											Mean, 31.0 ± 6.2 m
Wipfler B, 2011 [45]	1998–1999	Germany	DB-HT-auto	20	0	75.0	Mean, 30.0 ± 4.9		F: pins; T: metal staple or Bioscrew											Mean, 28.0 ± 6.7 m
			BPT-auto	31	3	61.2	Mean, 29.9 (range 25–55)	Mean, 2.6 m	F: press-fit fixation; T: Krackow											Mean, 105.6 (range 89–120) m
			4S-HT-auto	31	6	58.1	Mean, 34.2 (range 26–64)	Mean, 2.6 m	F: bottleneck; T: Mersilene tapes											
			4S-HT-auto	33*	4 (in total)	69.7	Mean, 27.0 ± 6.1	Range, 0.7–12 m	F: Endobutton; T: Acufex fixation post											24 m
Noh JH, 2011 [47]	2006–2008	Korea	BPTB-auto	28*		71.4	Mean, 26.0 ± 6.3		F: EndoButton; T: cannulated metallic interference screw											
			4S-HT-auto	33	9 (in total)	90.9	Median, 23 (range 20–51)	Mean, 1.7 ± 2.4 m	F: EndoButton CL; T: Intrafix and augmented with screw and washer											More than 24 m
			2S-NonIntra- Achilles-allo	32		81.3	Median, 22 (range 20–55)	Mean, 1.5 ± 1.9 m												
			4S-HT-auto	34	9	80.0	Mean, 26.1 ± 5.9	Range, 0.7–12 m	F: EndoButton; T: Acufex fixation post											Mean, 182.4 ± 7.2 m
Webster KE, 2015 [48]	1996–1998	Australia	BPTB-auto	31	9	72.7	Mean, 26.6 ± 6.7		F: EndoButton; T: cannulated metallic interference “Silk” screw											Mean, 183.6 ± 4.8 m
			4S-HT-auto	50	5	71.1	Mean, 29.0 ± 9.4	NA	F: BioScrew or EndoButton; T: BioScrew											More than 24 m
			4S-Contral- HT-auto	50	0	56.0	Mean, 29.5 ± 8.2													
			BPTB-auto	25	7	84.0	Mean, 31.0 ± 8.0	Mean, 15.0 ± 31.0 m	F&T: titanium interference screws											24 m
McRae S, 2013 [49]	2002–2010	Canada	QTB-auto	26	5	81.0	Mean, 30.0 ± 9.0	Mean, 21.0 ± 41.0 m	F: titanium interference screw; T: sheet (Intrafix)											
Lund B, 2014 [50]	2005–2009	Denmark																		

Table 2 (continued)

Eriksson K, 2001 [51]	1995– 1997	Sweden	BPTB-auto	50	8	64.5	Mean, 27.0 ± 6.0	NA	F&T: titanium interference screw F: Endobutton; T: titanium interference screws	6 m
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*Only the number of patients at the final follow-up was reported in the primary studies; The abbreviations as well as the full names for the grafts are listed in Table 1

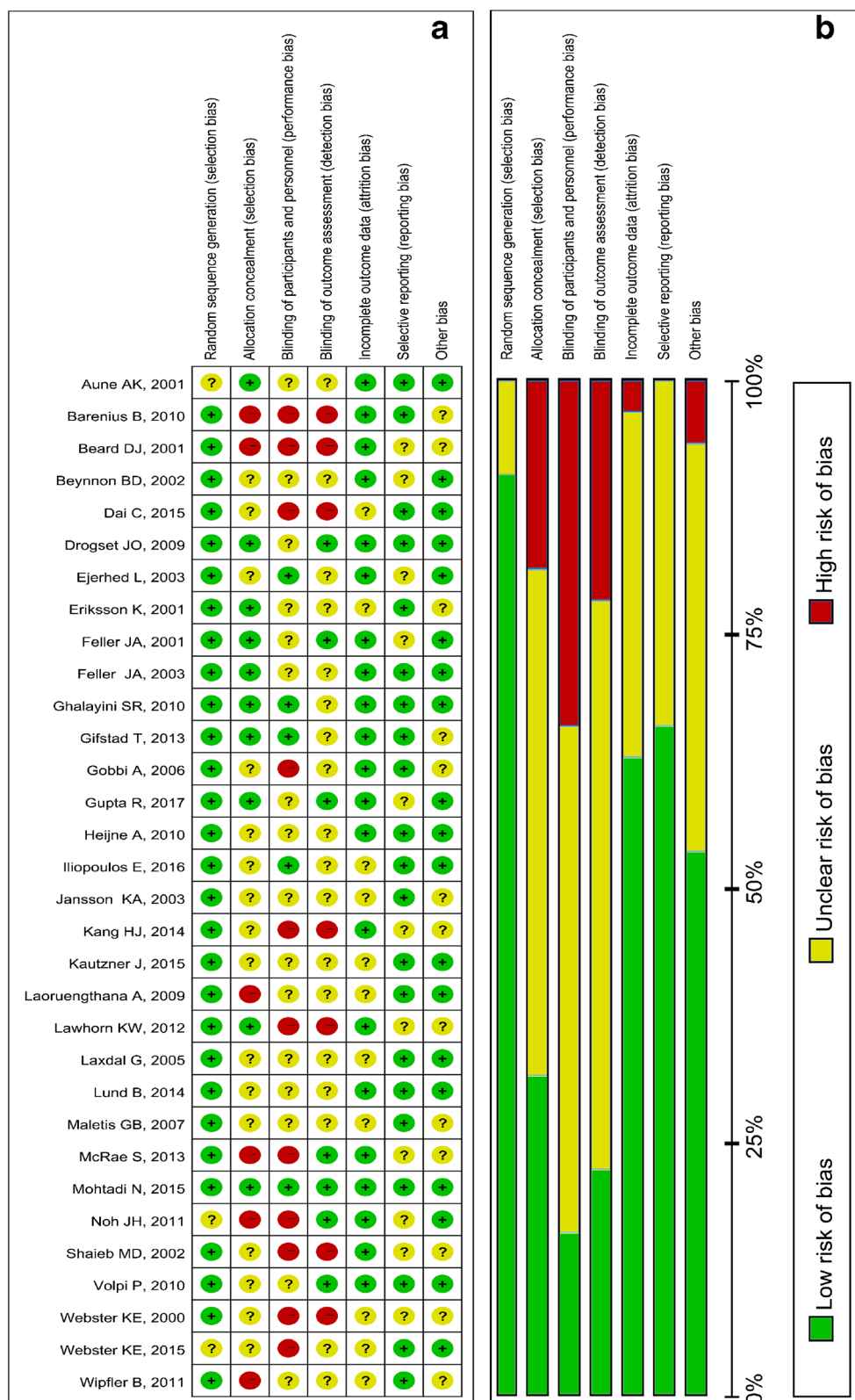
knee function, stability, complication, and re-rupture considerations of each graft was provided, offering a guidance that indicates which graft should be used.

Generally, reconstruction procedures using the BPTB or four-strand HT autografts are the most commonly discussed two treatments. Similar to observations from previous comparative studies, we confirmed that the BPTB autograft provides better knee stability than the four-strand HT autograft [52, 53]. Theoretically, as shown in former studies, the initial failure load of an evenly tensioned four-strand HT is approximately 4090 N, which is stronger than 2977 N for a 10-mm patellar tendon or 2160 N for an intact ACL [54–56]. This meets the principle consideration of strength for choice of grafts. From a biologic standpoint, however, it is well accepted that bone-to-bone healing (typically 4–6 weeks) is more accelerated than tendon-to-bone healing (typically 8–12 weeks) [57]. Recent research has also reported more fibrocartilage healing in the PT-bone interface while more fibrous healing in the HT-bone interface [58]. Therefore, the BPTB autograft is associated with more rigid healing both at the bone block-to-tunnel and the tendon section-to-tunnel interfaces. Additionally, when using suspensory fixation as the main approach to fix the HT graft, the distance between the articular surface and the point of fixation is increased, causing greater longitudinal (bungy effect) and transverse (windshield-wiper effect) motions. Therefore, a larger number of patients in the four-strand HT group were reported to have femoral tunnel widening [59].

As the least studied and used autografts, recent studies have shown excellent clinical results and low morbidity using the QTB autograft, especially when it is harvested as a free tendon without a bone block [60, 61]. In our results, the QTB autograft showed similar knee function and stability, while more of these patients reported satisfaction according to subjective IKDC score when compared with the BPTB, four-strand HT, and double-bundle HT autografts. The advantages of the QTB autograft include preservation of hamstring anatomy and function as well as less damage to the infrapatellar branch of saphenous nerve, thereby decreasing incidences of anterior knee pain and loss of sensitivity [62]. In addition, tendon defect after BPTB harvest may cause scar tissue, which leads to fat pad hypersensitivity and increased anterior knee pain when receiving a pressure load, such as during kneeling [63]. Though no more overall complications were related to BPTB, four-strand HT, and double-bundle HT autografts than the QTB autograft, complications of these grafts are generally more painful, resulting in an adverse effect on patients' subjective satisfaction. We therefore recommend that QTB autograft may be one of the best choices and should be used even over the BPTB and four-strand HT autografts.

In 1997, Kim et al. [64] proposed a new treatment option by preserving tibial insertions of HT when harvesting the graft. Since the blood supply of the tendon remains, the

Fig. 2 Risk of bias graph: the percentages of “high risk of bias,” “low risk of bias,” and “unclear risk of bias” for each item are presented in a bar diagram



healing of the tendon may be promoted, which allows rapid incorporation into the bone tunnel, minimizing risk of graft elongation. In addition, this technique helps preserve

mechanoreceptors in grafts [33, 57]. Our results fit well with former studies which show better anteroposterior stability for the four-strand HTB than for the four-strand HT autograft,

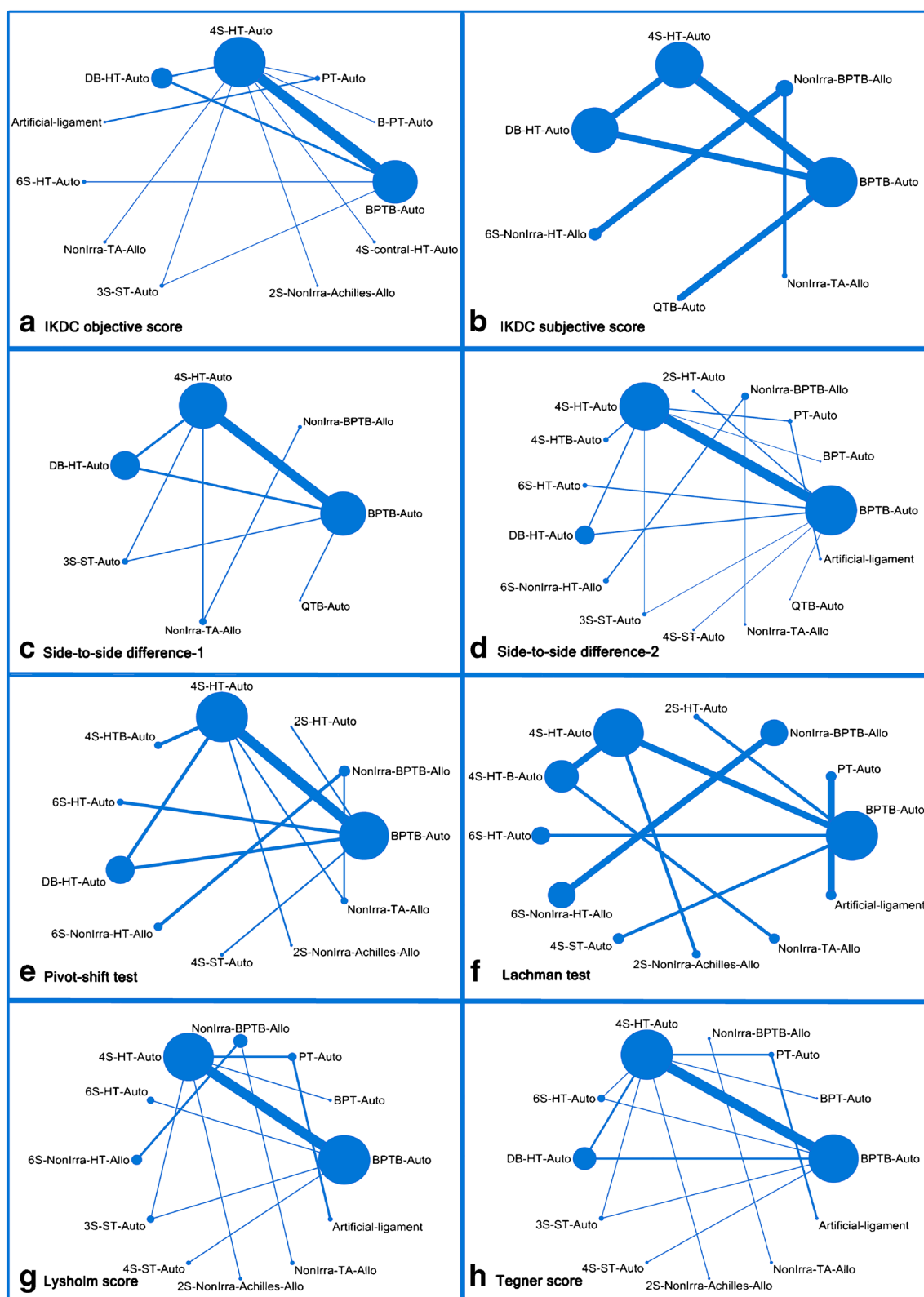
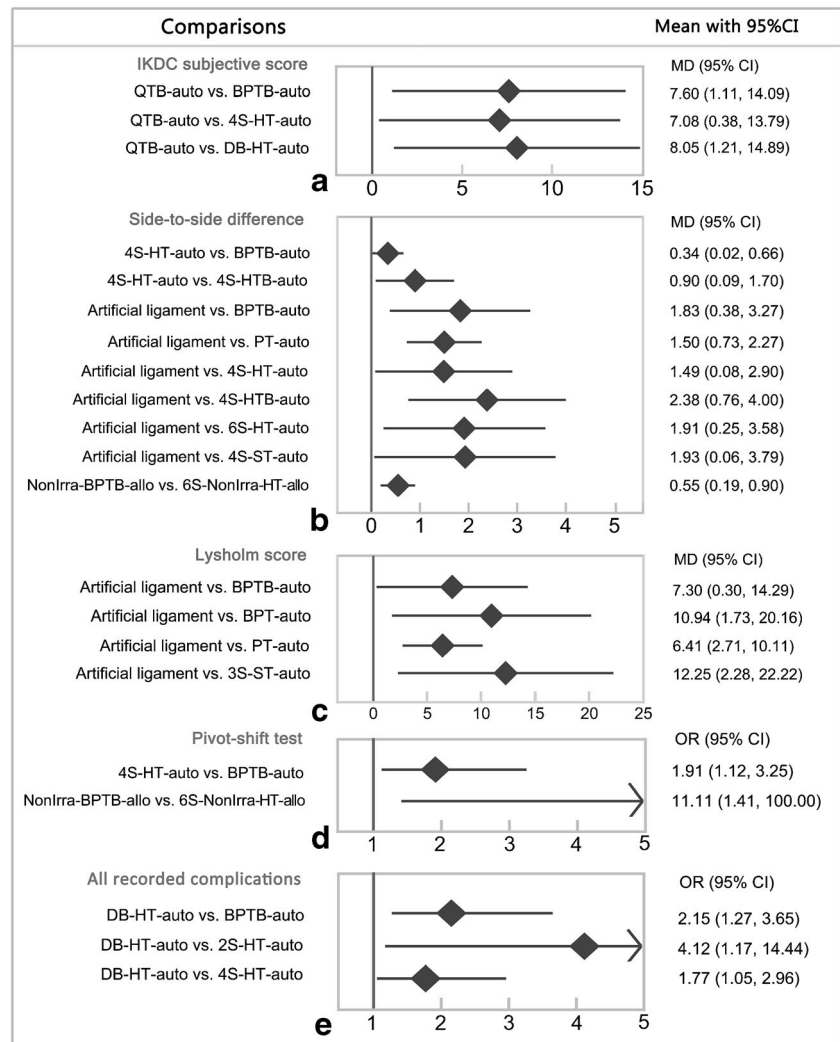


Fig. 3 Network plots for knee function and stability outcomes, including IKDC objective score (**a**), IKDC subjective score (**b**), side-to-side difference (reported as categorical data) (**c**), side-to-side difference (reported as continuous data) (**d**), pivot-shift test (**e**), Lachman test (**f**), Lysholm score (**g**), and Tegner score (**h**). Each node represents a treatment modality used

for ACLR, while each line represents a direct comparison between two treatments. The nodes and lines are weighted by the numbers of related patients and studies. Abbreviations as well as the corresponding full names of grafts are listed in Table 1

Fig. 4 Forest plots depicting the significant comparisons for each outcome in the network meta-analysis. No comparison was presented to be significant for IKDC objective score, side-to-side difference (continuous data), Lachman test, Tegner score, and re-ruptures. Abbreviations and full names of the grafts are listed in Table 1



indicating that the HTB autograft may be a better option for the reconstruction technique.

Many studies have focused on using allografts for ACLR following the advent of improved fixation device and tunnel techniques. The benefits of allografts include an absence of donor site morbidity, shortened surgical time, availability of large grafts, and no damage to the extensor and flexor mechanism [65]. However, to eliminate risks of rejection and disease transmission, treatments using cryopreservation and irradiation are often introduced to the grafts. These have been widely proved to be damaging to the biomechanical and histological properties [14, 15]. Recent meta-analyses have concluded that allografts are associated with less-than-acceptable knee outcomes when compared with autografts [66, 67]. However, in the subgroup of non-irradiated allograft, most studies reported a similar clinical outcome [67–69]. In our study, only a few non-irradiated allografts were available to be compared with autografts, and no difference in clinical outcome was found. We therefore suggest that non-irradiated allografts might be used as an alternative to autogenous

tendon grafts, while a conservative rehabilitation program should be advocated for allografts as the healing is relatively slow.

Artificial ligament is a synthetic material designed to provide a scaffold for the in-growth of neo-ligament and is related to a lack of donor site morbidity and shortened surgical time [70]. However, several histology and arthroscopy studies have suggested that artificial ligaments were not particularly useful as a scaffold-type graft, as they lack collagenization and neo-ligament in-growth [70, 71]. Chronic inflammatory responses have been reported in ruptured Leeds-Keio (LK) artificial ligaments to inhibit in-growth of the neo-ligament [72]. This may be why artificial ligaments were shown to provide inferior knee stability than almost all autografts. Thus, we do not suggest that artificial ligaments be selected for primary ACLR over other grafts, while they could be used for revision procedures if other options are not available, or used in combination with other grafts when carrying out complex ACLR. Additionally, to promote tendon graft integration and ligamentization following ACLR with artificial ligament,

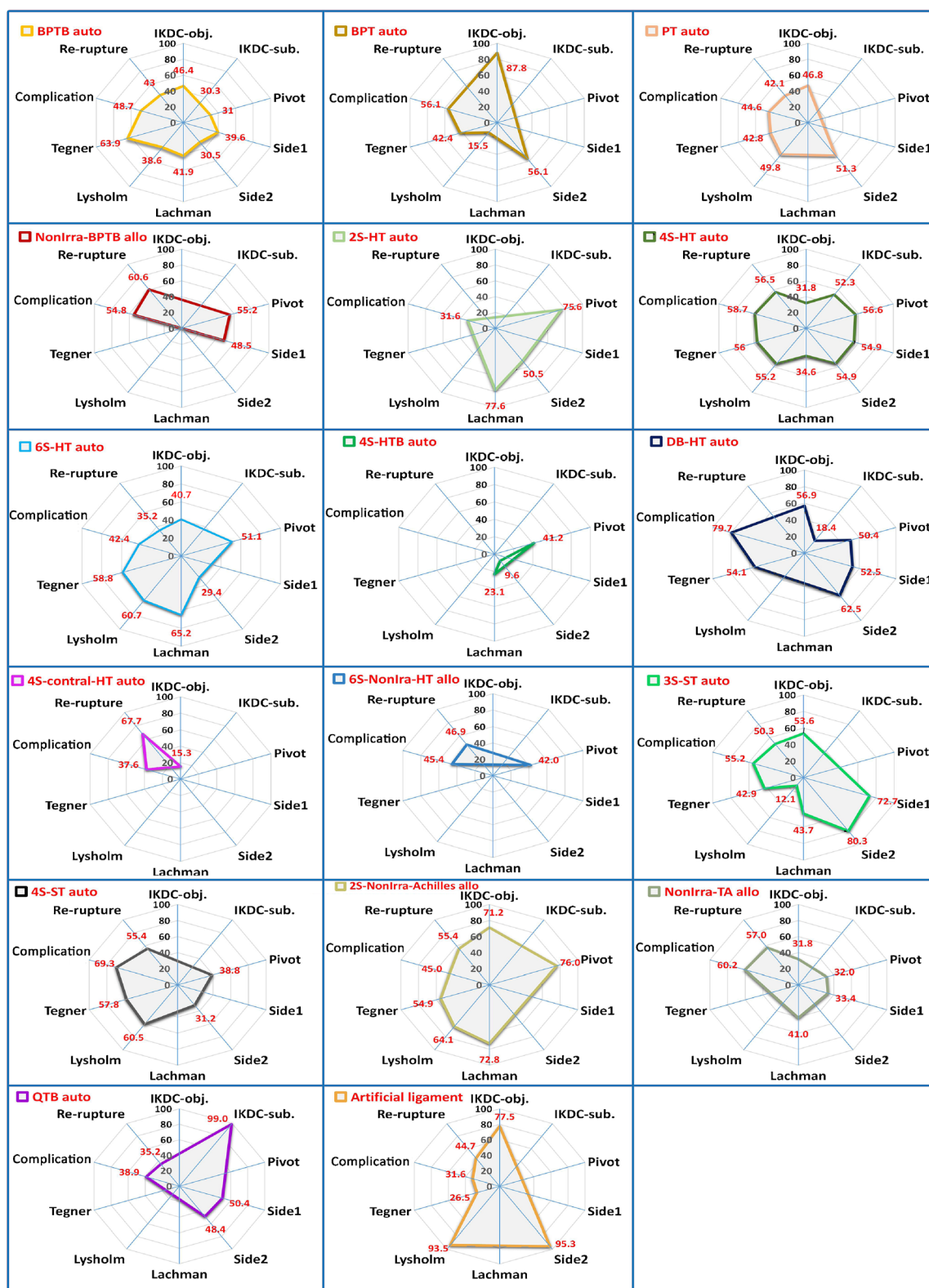


Fig. 5 Radar plots for the grafts with the axes presenting SUCRA values of the ten outcomes. IKDC-obj., IKDC objective score; IKDC-sub., IKDC subjective score; Pivot, pivot-shift test; Side1, side-to-side difference (categorical data); Side2, side-to-side difference (continuous data).

The SUCRA values are positively related to the magnitude of absolute measurement on the outcomes. Abbreviations and full names of the grafts are listed in Table 1

some biological agents or cells could be selected for augmentation of the reconstruction process [7, 73].

Limitations

Several weaknesses of this review must be stated. Firstly, available primary trials are predominately focused on comparing the BPTB and HT autografts, remaining a relatively small sample investigated other intervention arms. This may not allow for the detection of small differences between treatments, meaning there is risk of type II error. Secondly, different internal fixation devices and operation procedures were introduced for the various tendon transplants in the primary trials which have been recognized to confound treatment effects of different grafts themselves [74–76]. Finally, inconsistency on post-operative rehabilitation protocols may produce another potential risk of bias.

Conclusions

A recommendation for graft selection was proposed on the basis of the available direct and indirect evidence. We found that the four-strand HT autograft has an increased knee joint laxity than the BPTB autograft. The QTB autograft is recommended as one of the best choices and should be used even over the BPTB and four-strand HT autografts. We also recommend that non-irradiated allografts may be used as alternatives to autogenous tendon grafts. However, artificial ligaments are not recommended for primary ACLR, but could be used for revision procedures if other options are not available, or used in combination with other graft types.

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Compliance with ethical standards

Conflict of Interest The authors declare that they have no conflict of interest.

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